MySQL 8.1 Reference Manual

Version 8.1 Features and Functionality

1. Authentication and Security

MySQL 8.1 continues to use caching_sha2_password as the default authentication plugin but introduces enhancements to security:

```sql
-- Standard user creation with caching_sha2_password
CREATE USER 'username'@'host' IDENTIFIED BY 'password';

-- User with password history constraints
CREATE USER 'username'@'host' IDENTIFIED BY 'password'
PASSWORD HISTORY 5 -- Prevents reusing the last 5 passwords
PASSWORD REUSE INTERVAL 365 DAY; -- Cannot reuse a password within 365 days
```

Improved SSL/TLS connection security with TLS 1.3 recommended and older versions deprecated:

```sql
-- Check current TLS version
SHOW GLOBAL VARIABLES LIKE 'tls_version';

-- Configure secure connections
SET GLOBAL tls_version = 'TLSv1.2,TLSv1.3';
```

2. Important System Variables

Variable Name	Default	Description
max_connections	151	Maximum permitted number of client connections
innodb_buffer_pool_size	134217728	Size of the buffer pool for caching data and indexes
cte_max_recursion_depth	1000	Maximum recursion depth for common table expressions
innodb_adaptive_hash_index	ON	Whether the InnoDB adaptive hash index is enabled
create_admin_listener_thread	OFF	Create a thread for admin connections

3. SQL Syntax Elements

Enhanced JSON Functionality

MySQL 8.1 improves JSON path expressions with new syntax options:

```sql
-- Multiple path legs with wildcards and range operators
SELECT JSON_EXTRACT(data, '$.users[0 to 5].name') FROM app_data;

-- Using EXISTS operator with JSON_TABLE and NESTED PATH
SELECT *
FROM JSON_TABLE(
    data,
    '$.items[*]' COLUMNS(
        id INT PATH '$.id',
        name TEXT PATH '$.name',
        NESTED PATH '$.features[*]' COLUMNS(
            feature VARCHAR(50) PATH '$'
        )
    )
) AS jt
WHERE JSON_EXISTS(data, '$.tags[*] ? (@ == "premium")');
```

Lateral Derived Tables

Enhanced support for LATERAL JOIN operations:

```sql
SELECT customer.id, order_summary.total
FROM customers AS customer,
LATERAL (
    SELECT SUM(amount) AS total
    FROM orders
    WHERE orders.customer_id = customer.id
) AS order_summary;
```

Multi-Valued Indexes

Support for creating indexes on generated multi-valued columns from JSON arrays:

```sql
CREATE TABLE products (
    id INT PRIMARY KEY,
    data JSON,
    INDEX idx_tags ((CAST(data->'$.tags' AS JSON ARRAY)))
);
```

4. Deprecated Features

The following features are deprecated in MySQL 8.1 and will be removed in future versions:

- The binary client/server protocol version 4.0
- UTF8MB3 character set (use UTF8MB4 instead)
- TLS protocols older than TLS 1.2
- MyISAM storage engine for system tables
- Creation of tables without a PRIMARY KEY
- The FLOAT and DOUBLE types without a specified precision

5. InnoDB Enhancements

- Improved parallel threads for online DDL operations
- Enhanced foreign key constraint validations
- Better performance for secondary index lookups
- Faster shutdown and startup times
- Improved temporary table performance
- Enhanced crash recovery with parallel threads

6. Error Codes

New error codes in MySQL 8.1:

- 5001: JSON schema validation failed
- 5120: Parallel DDL operation aborted
- 5200: Table encryption policy violation
- 5310: Temporary table size limit exceeded
- 5400: Resource group operation failed

7. Performance Schema Enhancements

- New memory instrumentation for improved memory tracking
- Enhanced lock instrumentation
- More detailed transaction monitoring
- Thread pool instrumentation

8. Optimizer Improvements

- Improved cost model for complex JOIN operations
- Enhanced histogram statistics for better query plans
- Subquery materialization optimizations

- More efficient execution for window functions
- Improved handling of derived tables
- Histogram-based statistics for better cardinality estimation

9. Character Set and Collation

- UTF8MB4 is still the default character set
- New specialized collations for language-specific sorting
- Improved case-insensitive operations for performance
- Deprecated utf8mb3 with planned removal in future versions